

MAIN ARTICLE - REMOVED ORIGINAL CONTENT

improve the electronic communication in quasi-2D perovskites.

According to the density functional theory (DFT) calculation, the IDMP has a higher dipole moment (3.54 D) than that of AMP (1.60 D), indicating the stronger interaction between the IDMP ligand and the lead bromide ($[\text{PbBr}_6]^{4-}$) octahedra. The adsorption modes for IDMP on the perovskite surface include single-anchor and double-anchor types, while the AMP can show only single-anchor on the perovskite surface (Supplementary Fig. 5). Their corresponding formation energies are summarized in Fig. 1c, testifying their probability of existence and the highest probability for creating double-anchor connection between IDMP and the perovskite surface. Differential charge density analyses further reveal the formation of stronger chemical coordination bonds between IDMP and the perovskite surface (Supplementary Fig. 6). Bader charge analysis reveals that the charge transfer values for AMP- and IDMP-treated perovskites are 0.09 e and 0.29 e, respectively, suggesting that the IDMP ligand has stronger capability to transport charge carriers into perovskite emitters, promising a more efficient interlayer charge transfer.

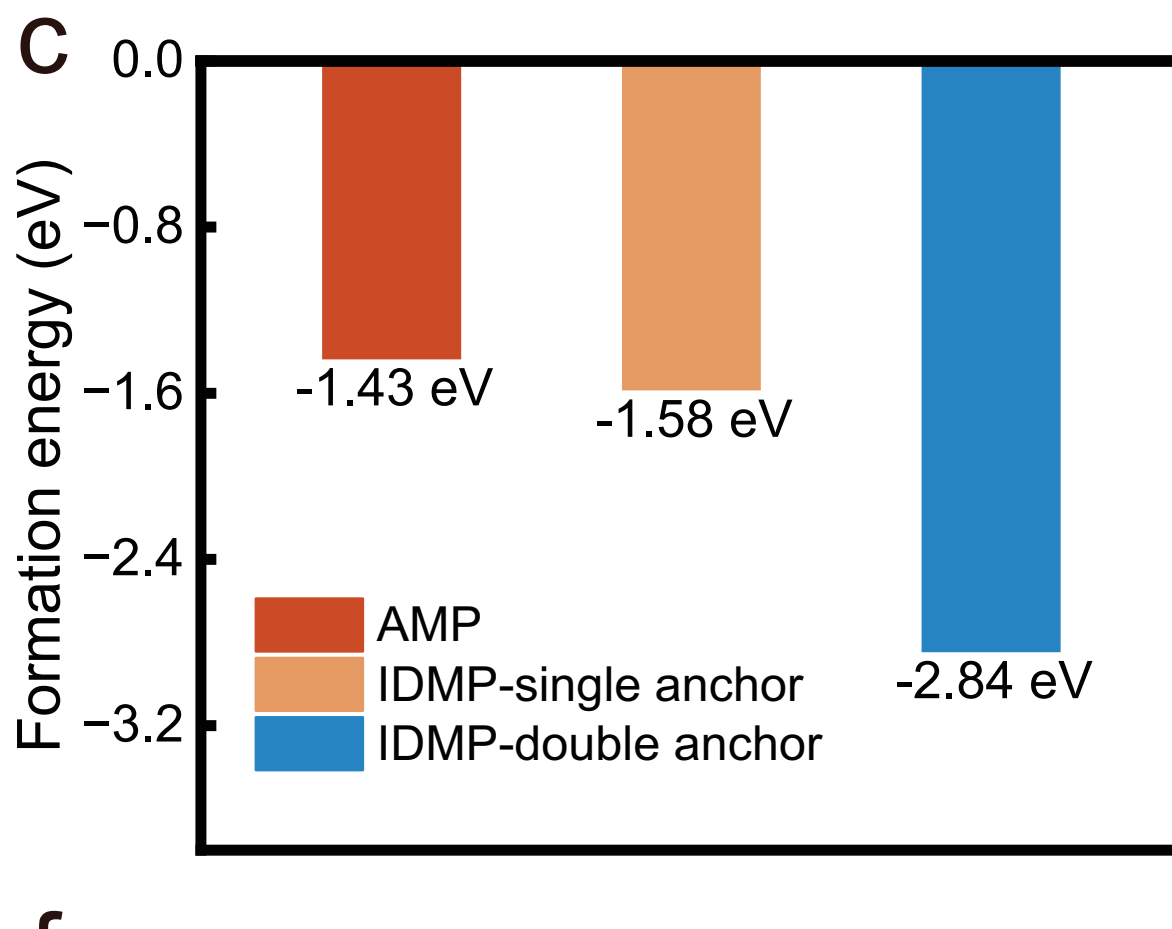


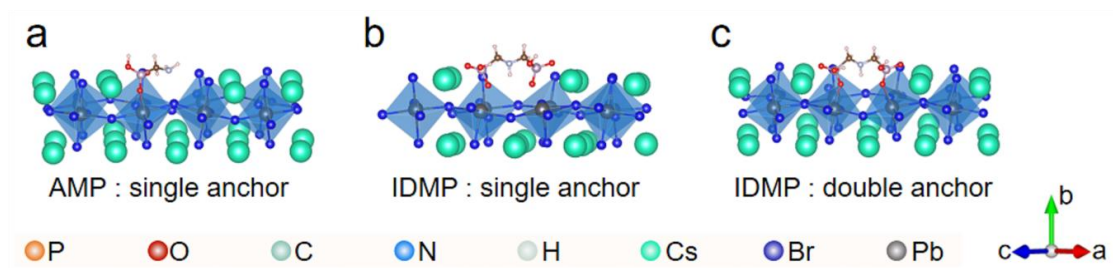
Fig. 1 | Regulation of exciton behaviors at interlayer and intra. **a** Molecular structure and calculated electrostatic surface potential map of iminodi(methylphosphonic acid) (IDMP) ligand. **b** The dielectric constant (left axis) and the corresponding exciton binding energy (right axis) for pristine, (aminomethyl) phosphonic acid (AMP) -treated, and IDMP-treated perovskite films, respectively. **c** The formation energies of AMP and IDMP ligands on the surface of perovskite simulated with the PbBr_2 terminated model. The IDMP ligand can anchor with the perovskite through single- and double-ends. **d** Steady-state photoluminescence (PL) spectra, **e** time-resolved PL spectra, and **f** PL quantum yield (PLQY) for pristine, AMP-treated, and IDMP-treated perovskite films, respectively. **g** Schematic illustration of the working mechanism of the IDMP ligand.

SUPPLEMENTARY INFORMATION - REMOVED ORIGINAL
CONTENT

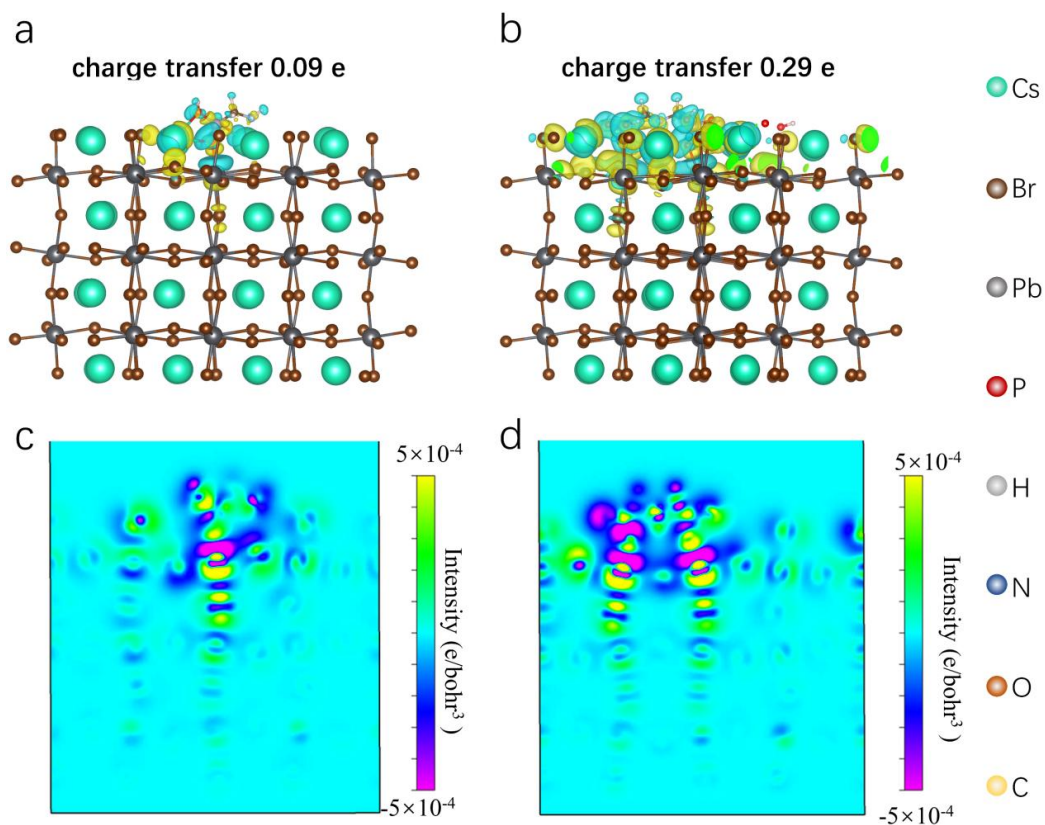
Supplementary Fig. 5 | Absorption modes.

Supplementary Fig. 6 | Calculated charge density and electron localization fu

Supplementary Fig. 16 | SCLC measurements and the DOS calculations.

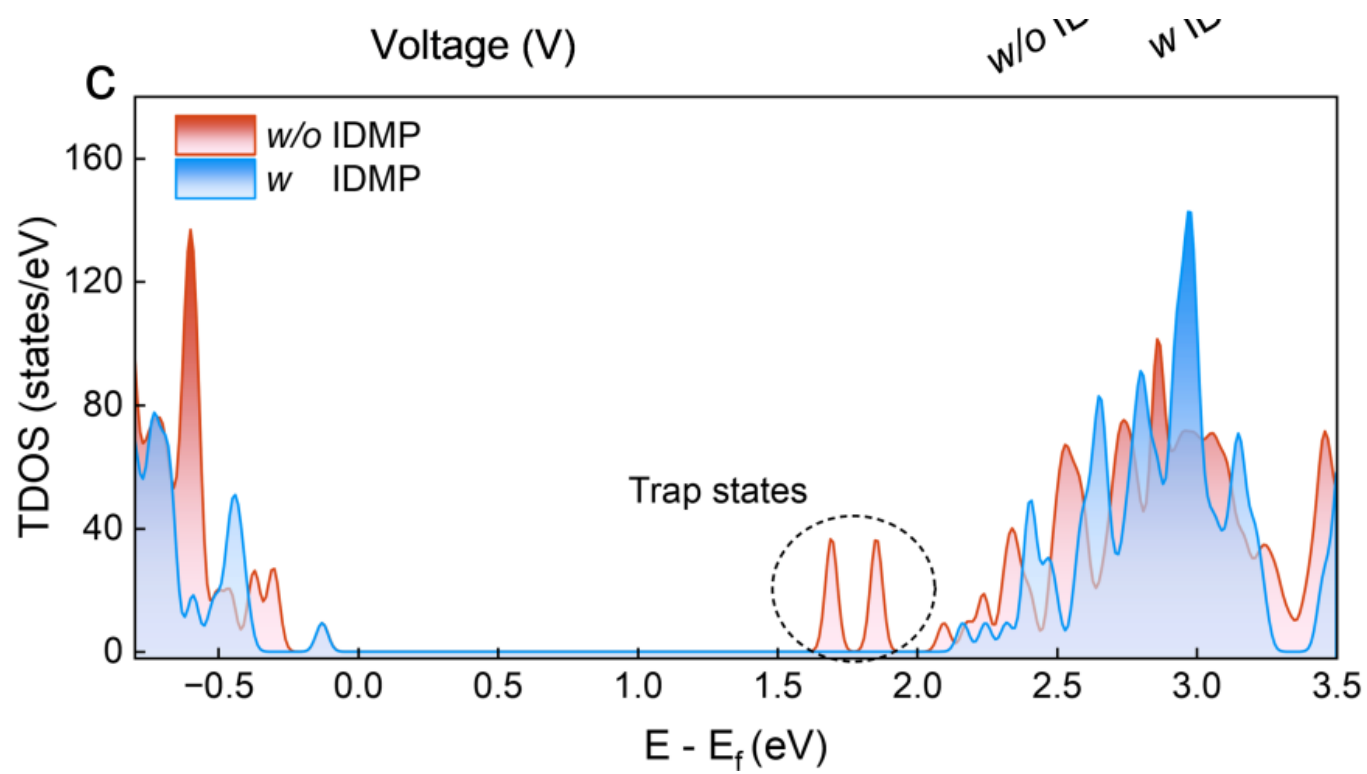


Supplementary Figure 5 | Absorption modes. The absorption modes for **a** AMP, **b, c** IDMP on the perovskite surface.



Supplementary Figure 6 | Calculated charge density and electron localization function images.

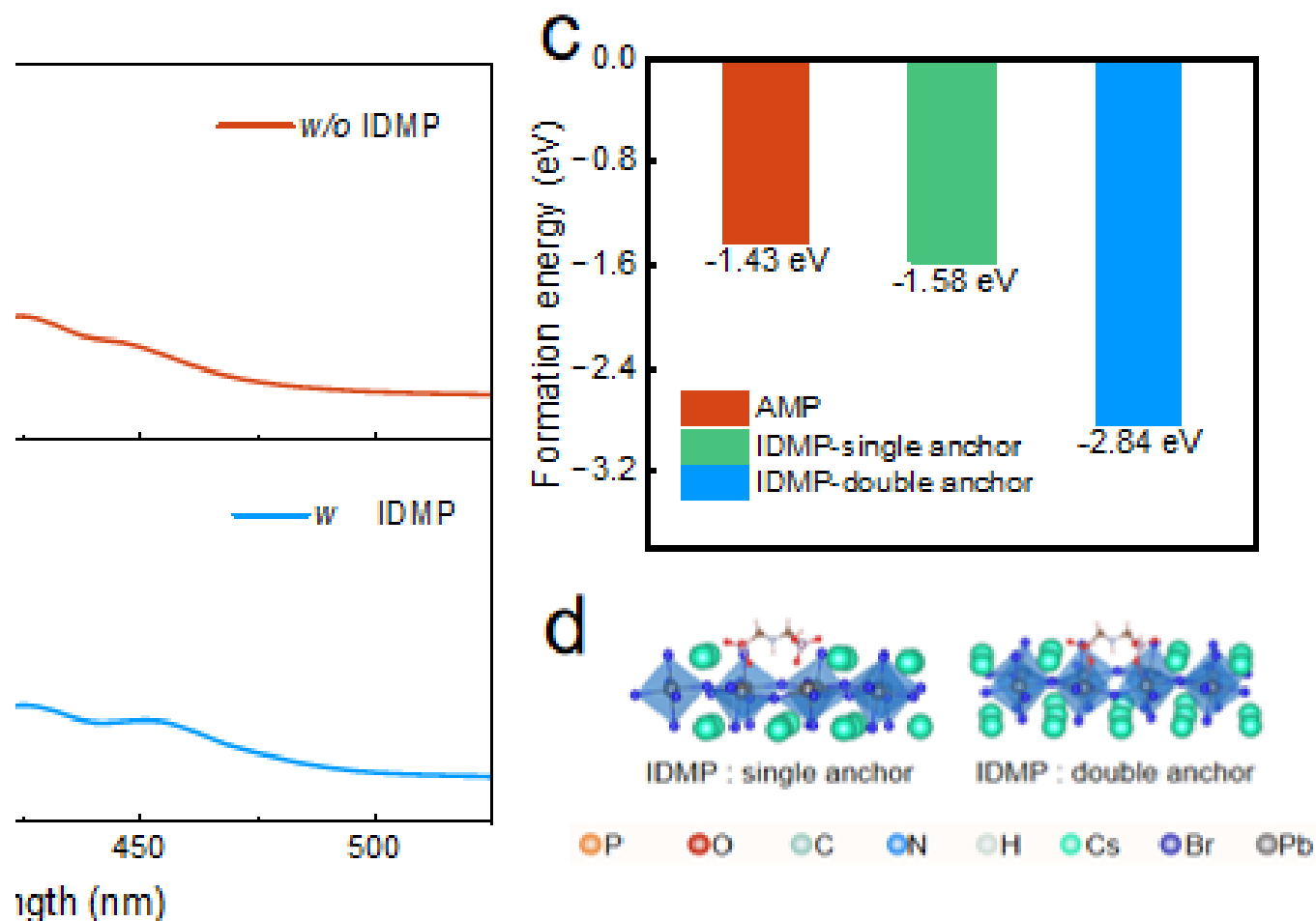
Calculated charge density differences for **a** AMP- and **b** IDMP-treated perovskite films. Electron localization function images of perovskite films with **c** AMP and **d** IDMP.



Supplementary Figure 16 | SCLC measurements and the DOS calculations. **a** Space charge limited current measurements of hole-only devices based on pristine and IDMP-treated perovskite films. The structure of hole-only devices was ITO/PEDOT:PSS/PVK:PBD/2PACz/perovskites/MoO₃/Al. **b** The corresponding calculated defect state densities. **c** Electronic density of states for pristine and IDMP-treated perovskite films. The error bar is $\pm 0.4 \times 10^{17} \text{ cm}^{-3}$. The defect density decreased from $4.6 \pm 0.4 \times 10^{17} \text{ cm}^{-3}$ in the pristine perovskite film to $1.0 \pm 0.4 \times 10^{17} \text{ cm}^{-3}$ in the IDMP-treated perovskite film. The CsBr-terminated surface of the pristine perovskite film exhibits distinct peaks at approximately 1.6 eV and 1.9 eV above the Fermi level, which stem from the deep-level defect states in the materials. With the IDMP treatment, these signals are eliminated, indicating the removal of deep-level defects.

RESPONSE LETTER - REMOVED ORIGINAL CONTENT

In theoretically, on the one hand, according to the molecule structures, the space steric hindrance for short-chain IDMP molecules is much smaller than those of larger organic spacer cations of PBA^+ and MOPA^+ containing aromatic rings, which plays the dominated roles of intercalating the 3D perovskite into slices (as we described in the above Q1). On the other hand, we have used density functional theory (DFT) calculations to evaluated the adsorption modes of IDMP on perovskites (**Fig. R2c-d**), which well supports the claimed double-end anchoring mode and meanwhile, helps us exclude the contribution of IDMP molecules to the steric hindrance in the system. Thanks.



t concentrations of IDMP-treated perovskite
tra for pristine and IDMP-treated perovskite
gands on the surface of perovskite simulated
odes for IDMP on perovskite surface.

As for more evidence of suppression of deep-level defects in the perovskite, we performed density functional theory (DFT) calculations. According to the electronic density of states (DOS) in **Fig. R5**, the CsBr-terminated surface of the pristine perovskite film exhibits distinct peaks at approximately 1.6 eV and 1.9 eV above the Fermi level, which stem from the deep-level defect states in the materials (*Nature*, 2024, 631, 73-79). With the IDMP treatment, these signals are eliminated, which indicates the removal of deep-level defects.

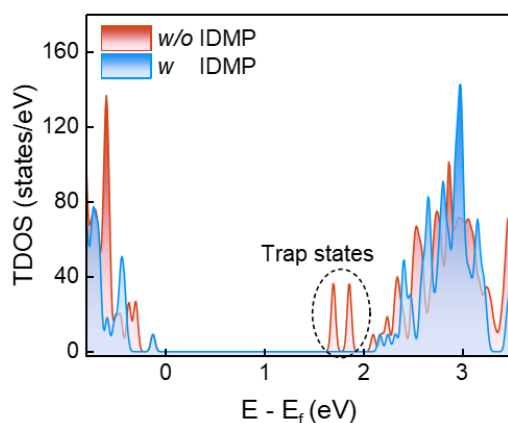


Figure R5. (Supplementary Figure 16c in revised Supplementary Information) Electronic density of states (DOS) for pristine and IDMP-treated perovskite films.

As shown in Fig. R15-16, we have added the coordinate axes and marked the elements represented by each color in our revised Supporting Information. Thanks.

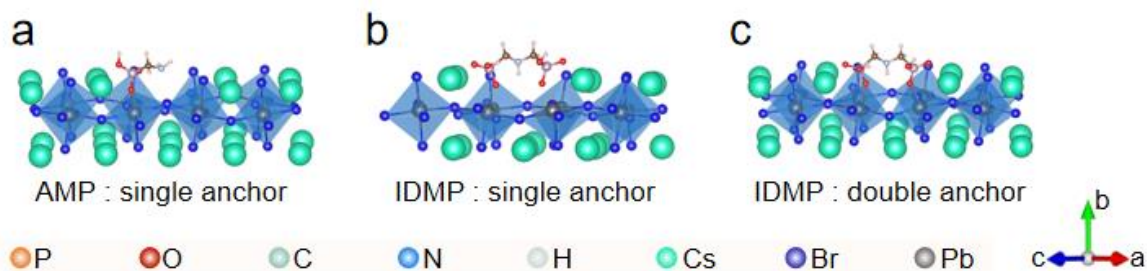


Figure R15 (Supplementary Figure 5 in revised Supporting Information). The absorption modes for (a) AMP, (b-c) IDMP on the perovskite surface.

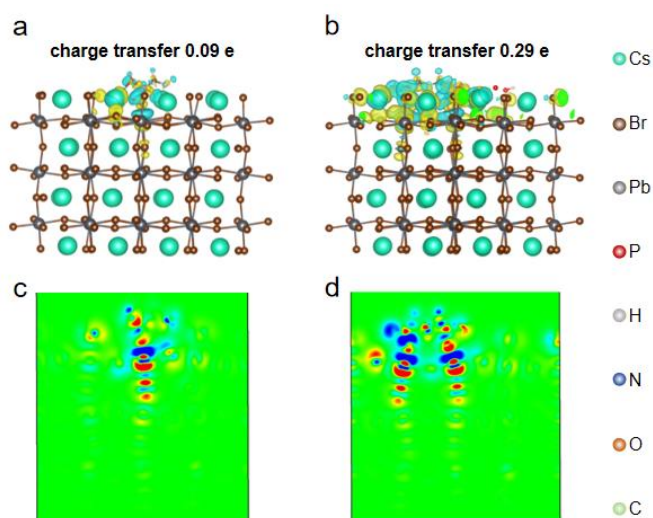


Figure R16 (Supplementary Figure 6 in revised Supporting Information). Calculated charge density differences for (a) AMP- and (b) IDMP-treated perovskite films. Electron localization function images of perovskite films with (c) AMP and (d) IDMP.